HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Analyzed student demographic data to identify participation trends for **socio-economically disadvantaged backgrounds**, **mature students**, and **refugee applicants**, informing targeted outreach strategies.
- Developed and maintained **Power BI dashboards** visualizing key performance indicators for student access, retention, and progression, enabling data-informed decision-making for the Access Office.
- Extracted and transformed student data from disparate sources using **Python (pandas)** and **SQL** to prepare datasets for analysis and reporting on **National Access Plan** target groups.
- Collaborated with the **Institutional Research and Data Office** to gather and validate statistical reports, ensuring data accuracy for university strategy and benchmarking initiatives.
- Designed and implemented data collection methodologies to support **impact measurement and reporting** for widening participation programmes, contributing to evidence-informed policy.
- Performed ad-hoc data analysis using **R** to investigate student engagement patterns and identify factors influencing retention for specific demographic cohorts.
- Supported the **Maynooth Access Programme (MAP)** by providing timely and accurate data insights, contributing to the service's capacity for monitoring and reporting.

SKILLS

Data Analysis & BI – Power BI, Tableau, R, pandas, NumPy, DAX, Power Query, statistical analysis, exploratory data analysis, A/B testing, time-series analysis, data modelling, data warehousing, business intelligence

Data Handling & ETL – SQL, PostgreSQL, MySQL, Oracle, Python scripting, data pipelines, ETL

Programming & Scripting – Python, Java, Spring Boot, REST APIs, Flask, FastAPI

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, shell scripting, cron jobs

Project Management & Collaboration – Jira, Confluence, Agile, Scrum ceremonies, change-control management, cross-functional collaboration

Domain Knowledge – Widening participation, student access, retention, progression, National Access Plan, higher education strategy, regulatory awareness, GDPR

PROJECTS

MSc Applied Data Analytics Project: Student Success Predictor

- Developed a **Python-based predictive model** using scikit-learn to identify students at risk of non-progression, analyzing features from academic records and engagement data.
- Extracted and cleaned a dataset of **~15,000 student records**, performing feature engineering to capture key indicators of academic performance and support utilization.
- Evaluated model performance using **AUC and F1-score**, achieving a **~0.82 AUC** on a held-out test set and providing actionable insights for proactive student support interventions.

HCL Financial Services Dashboard Enhancements

- Enhanced a **payments operations dashboard** by developing new Python scripts to automate the generation of daily reconciliation reports, reducing manual effort by approximately **6 hours per week**.
- Wrote complex **SQL queries** against PostgreSQL to extract and aggregate transaction data, supporting operational alerting and monitoring workflows for a financial services client.
- Collaborated with business analysts to refine data visualizations and reporting logic within the dashboard, improving clarity for **operations teams** on failed transaction flows.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)